

ALEX ARMENGOL URPI

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EDUCATION

Massachusetts Institute of Technology

PhD in Human-Computer Interfaces

- “La Caixa” Fellowship for studies at MIT (full tuition and stipend)

Cambridge, MA

2018 – June 2023

Massachusetts Institute of Technology

M.S., Mechanical Engineering.

- Concentration on Human-Computer Interfaces for IoT and AR/VR

Cambridge, MA

2016 - 2018

CFIS - Universitat Politècnica de Catalunya

Dual Major Bachelor's Degree:

- Telecommunications Engineering.
- Industrial Engineering.

- CFIS scholarship fellow to pursue 2 major engineering degrees simultaneously. 40 admitted students/year in Spain.

Barcelona, Spain

2011 - 2016

RESEARCH EXPERIENCE

MIT, Department of Mechanical Engineering

Research Scientist

Cambridge, MA

October 2025 - Present

- Developed a transformer-based open-vocabulary 3D grounding system for real-time object localization in AR environments. Fused point-cloud geometry with DINOv2 visual embeddings in a PyTorch architecture to enable zero-shot localization of unseen objects. Leveraged SLAM camera poses to improve accuracy and runtime efficiency in cluttered scenes. Comparable results to closed-set, fully supervised models.
- Built a real-time AR guidance system that uses LLMs to generate free-form 3D overlays from geometric primitives. Integrated depth camera perception and AI-based object tracking to dynamically adapt LLM-generated overlays to object geometry, task context, and headset pose, enabling flexible spatial instructions beyond predefined AR guidance templates.

MIT, Department of Mechanical Engineering

Postdoctoral associate

Cambridge, MA

June 2023 – October 2025

- Developed a multimodal sensing framework to capture tacit knowledge from expert demonstrations in real-world tasks (crafts and sports). Instrumented experts, tools, and environment with sensors including eye tracking, EMG, accelerometers, airflow sensors, and scene cameras to capture behavioral and biomechanical signals. Built a data pipeline to generate instructional videos and AI-ready datasets for modeling differences between experts and novices.

MIT, Department of Mechanical Engineering

Research Assistant. Auto-ID Lab. Advisor: Sanjay E. Sarma

Cambridge, MA

2015 – June 2023

- Study of human implicit learning and tacit knowledge to enhance human expertise and AI performance:
 - Enhancing AI systems: Designed a conv–deconv neural net architecture to predict human gaze distributions, demonstrating that AI models benefit from tacit human knowledge signals beyond labeled data. Our system outperformed by 20+ points baseline image classification models when data is scarce.
 - Enhancing human performance: Identified gaze and attention patterns as markers of tacit expertise and generated expert attention heatmaps from eye-tracking data to guide learners, improving performance in visual tasks of trained individuals compared to an equally trained control group.
- Development of novel interfaces for foveal (overt) and peripheral (covert) visual attention tracking using EEG
- Brain-Computer Interfaces for Virtual Reality and the Internet of Things.
- Development of a thermal comfort sensing scheme using wearables for physiological signal monitoring.
- Research in RFID tag antenna-based sensing and in anti-counterfeiting architectures for the IoT.

INDUSTRY EXPERIENCE

LIFEVERSE AI

Part time consultant

London, UK

Spring 2023

- Consulting on research directions on AI for behavioral understanding on children's attention for education

PHILIPS NORTH AMERICA

Augmented Reality Research Intern.

Cambridge, MA

Summer 2019

- Research on Augmented Reality-guided surgical procedures.

ALMATECH INC.
Co-leader of engineering department.

- Worked on UPC-derived spinoff company devoted to the WPT domain.

Barcelona, Spain
Oct 2014 – March 2015

KEONN TECHNOLOGIES
Intern. System integration.

- Involved in an RFID interactive retail system project.

Barcelona, Spain
Summer 2014

SELECTED PUBLICATIONS

- [1] *HoLocate: An Open-Vocabulary, Zero-Shot 3D Grounding Framework for Augmented Reality*
Alex Armengol-Urpi, Dulanga Weerakoon, Andres Salazar, Sanjay Sarma
In preparation.
- [2] *Free Form 3D Overlays for Open-ended Augmented Reality Task Assistance Using LLMs*
Alex Armengol-Urpi, Dulanga Weerakoon, Andres Salazar, Sanjay Sarma
In preparation.
- [3] *Cognitive Reinforcement: Capturing Tacit Knowledge and Enhancing Expertise with a Biofeedback Interface for Visual Attention*
Alex Armengol-Urpi, Andres Salazar, Pawan Sinha, Sanjay Sarma
Journal of Neural Engineering. March 2026.
- [4] *Brain-Hack: Remotely Injecting False Brain-Waves with RF to Take Control of a Brain-Computer Interface*
Alex Armengol-Urpi, Reid Kovacs, Sanjay Sarma
Workshop on CPS&IoT Security and Privacy. **Best paper award.**
- [5] *Brainwave-Augmented Eye Tracker: High-Frequency SSVEPs Improves Camera-Based Eye Tracking Accuracy*
Alex Armengol-Urpi, Andres Salazar, Sanjay Sarma
IUI 2022.
- [6] *A system approach for closed-loop assessment of neuro-visual function based on convolutional neural network analysis of EEG signals*
Simon Stock, **Alex Armengol-Urpi**, Bálint Kovács, Heiko Maier, Marius Gerdes, Wilhelm Stork, Sanjay Sarma
SPIE Neurophotonics 2022
- [7] *Towards Industrial IoT-AR Systems Using Deep Learning-Based Object Pose Estimation*
Yongbin Sun, Sai Nithin Reddy Kantareddy, Joshua Siegel, **Alex Armengol-Urpi**, Xiaoyu Wu, Hongyu Wang, Sanjay Sarma
IPCCC 2019
- [8] *Magichand: Interact with IoT Devices in Augmented Reality Environment*
Yongbin Sun, **Alex Armengol-Urpi**, Sai Nithin Reddy Kantareddy, Joshua Siegel, Sanjay Sarma
IEEE VR 2019
- [9] *Sublime: a Hands-Free Virtual Reality Menu Navigation System Using a High-Frequency SSVEP-based Brain-Computer Interface*
Alex Armengol-Urpi, Sanjay Sarma
VRST 2018

Patents

- [10] Sanjay E Sarma, Rahul Bhattacharyya, Joshua E Siegel, **Alex Armengol-Urpi**, Pankhuri Sen, Sai Nithin Reddy Kantareddy. “Antenna and system for wireless sensing of health monitoring”
- [11] Sanjay E Sarma, **Alex Armengol-Urpi**. “Brain-computer interface for user's visual focus detection”

FELLOWSHIPS, AWARDS & HONORS

- **MIT News feature.** Research on tacit knowledge and visual attention highlighted by *MIT News*, 2026.
- **Best paper award.** Workshop on CPS&IoT Security and Privacy, 2023.
- **“La Caixa” Fellowship.** Full funding for 2 years of PhD program at MIT, 2018.
- **CFIS Fellow.** Telecom. Engineering Degree tuition covered by the Technical University of Catalonia, 2011.
- **Youth Research Awards.** Award given by the Government of Catalonia for my high school final thesis, 2010.
- **Gov. of Catalonia Fellow.** 1st year college tuition covered by Gov. of Catalonia for Honors in high school, 2010.
- **Catalunya Caixa Bank Fellow.** For outstanding score in the National University entry exam (13.48/14), 2010.